

EVOLINE-VILLAZ, Switzerland

WMO: 067220

Lat: 46.1122N

Lon: 7.5086E

Elev: 5997

StdP: 11.78

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.2	10.6	-15.6	2.9	14.0	-9.7	4.1	16.6	14.7	28.0	11.4	30.0	3.0	40	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	72.5	55.8	69.5	54.9	66.9	53.8	57.4	69.0	56.0	66.9	54.8	64.8	6.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.8	74.3	60.5	51.4	70.5	59.8	50.1	67.2	59.3	27.3	69.2	26.4	67.2	25.5	65.0	62.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.3	8.8	3.5	77.2	5.2	3.2	-0.2	79.5	-3.3	81.4	-6.2	83.2	-9.9	85.5		
(5)			1.1	60.3	4.2	1.1	-1.9	61.1	-4.3	61.8	-6.7	62.5	-9.7	63.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	26.8	26.5	32.4	39.4	44.7	53.0	56.0	55.2	49.2	44.0	34.6	29.8		
	DBStd	12.85	7.90	8.85	8.06	7.22	7.03	7.04	6.30	6.70	6.40	7.50	8.46	8.08		
	HDD50	3913	718	658	544	322	193	48	21	27	90	205	462	625		
	HDD65	8750	1183	1078	1009	768	629	362	284	311	475	649	912	1090		
	CDD50	651	0	0	0	4	28	139	207	187	65	20	1	0		
	CDD65	13	0	0	0	0	0	3	4	6	0	0	0	0		
	CDH74	50	0	0	0	0	0	15	16	19	0	0	0	0		
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0		
	WSAvg	3.3	2.7	2.7	3.4	4.2	4.0	4.0	3.9	3.5	3.1	3.0	2.7	2.7		
	PrecAvg	27.8	2.0	1.9	1.9	1.9	2.8	2.6	2.6	2.7	2.0	2.2	2.6	2.4		
(15) Precipitation	PrecMax	36.2	5.8	8.1	4.4	3.9	6.3	5.1	6.1	5.6	6.1	6.6	7.3	6.2		
	PrecMin	17.3	0.4	0.1	0.6	0.1	0.9	1.2	0.8	1.0	0.6	0.2	0.7	0.2		
	PrecStd	4.5	1.4	1.8	1.0	1.0	1.7	1.1	1.0	1.1	1.3	1.5	1.7	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	49.0	53.1	60.4	66.9	75.6	76.1	76.9	67.9	63.1	54.7	48.0		
		MCWB	35.0	35.6	38.8	44.7	51.3	56.3	57.3	56.0	53.3	47.9	40.7	34.8		
	2%	DB	43.2	44.7	49.4	56.6	62.5	71.3	72.2	71.8	64.7	59.0	50.8	44.5		
		MCWB	33.1	33.1	36.9	42.6	48.5	55.4	55.5	55.9	51.8	45.9	38.8	33.6		
	5%	DB	40.5	41.1	46.4	53.0	59.0	67.8	69.3	68.4	61.5	56.4	47.7	42.2		
		MCWB	31.8	31.6	35.4	41.1	47.0	54.2	54.8	54.6	50.5	44.8	37.8	32.3		
	10%	DB	37.8	37.7	43.3	49.9	55.6	64.4	66.4	65.4	58.5	53.5	45.1	39.9		
		MCWB	30.5	29.9	33.5	39.3	45.2	52.7	54.0	53.6	48.9	43.6	36.3	31.1		
	0.4%	WB	37.7	37.0	40.7	45.9	52.5	58.4	58.9	59.1	55.2	50.2	43.4	37.7		
		MCDB	42.2	44.3	49.1	57.5	63.4	71.6	70.9	71.2	64.9	58.3	49.1	42.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.1	34.8	38.1	43.5	50.0	56.7	57.2	57.2	53.1	48.1	40.8	35.8		
		MCDB	40.4	41.9	46.3	54.4	60.4	68.4	69.1	68.2	62.0	55.4	47.7	41.3		
	5%	WB	33.1	32.6	36.4	41.8	47.9	55.0	55.9	55.8	51.4	46.5	39.1	34.2		
		MCDB	39.0	39.7	44.2	51.4	56.4	65.7	66.6	66.0	59.6	53.4	45.7	39.8		
	10%	WB	30.9	30.4	34.6	40.1	46.2	53.3	54.6	54.4	49.8	44.9	37.3	32.3		
		MCDB	37.2	37.3	41.8	48.6	54.1	62.9	64.6	63.8	56.7	51.9	43.8	38.2		
	5% DB	MDBR	9.7	10.6	12.7	14.5	14.4	15.4	16.0	14.4	13.6	12.8	10.5	10.0		
		MCDBR	10.9	12.1	15.3	17.4	18.0	18.7	18.9	17.6	16.2	14.9	12.6	11.2		
		MCWBR	7.9	8.2	9.2	10.1	10.4	10.2	9.6	9.7	9.8	9.8	8.3	7.2		
		MCWBR	10.4	11.7	14.0	16.4	16.3	17.2	17.2	15.9	14.5	13.4	11.8	10.9		
(35) Clear-Sky Solar Irradiance	taub		0.226	0.248	0.289	0.329	0.346	0.351	0.347	0.336	0.316	0.293	0.256	0.225		
		taud	2.326	2.270	2.214	2.231	2.292	2.319	2.324	2.367	2.418	2.460	2.437	2.358		
	Ebn at Noon		287	298	298	294	292	290	289	289	286	279	272	275		
		Edh at Noon	23	30	38	41	40	40	39	36	31	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		471	747	1097	1393	1590	1893	1895	1650	1290	836	483	388		
	RadStd		54	100	148	185	136	164	153	131	106	91	69	56		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (104 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	+1.28	+0.72	+0.54	N/A	N/A	+101
										+29

Nomenclature: See separate page